

Low protein intake: the impact on calcium and bone homeostasis in humans.



Increasing dietary protein results in an increase in urinary calcium. Despite over 80 y of research, the source of the additional urinary calcium remains unclear. Because most calcium balance studies found little effect of dietary protein on intestinal calcium absorption, it was assumed that the skeleton was the source of the calcium. The hypothesis was that the high endogenous acid load generated by a protein-rich diet would increase bone resorption and skeletal fracture. However, there are no definitive nutrition intervention studies that show a detrimental effect of a high protein diet on the skeleton and the hypothesis remains unproven. Recent studies from our laboratory demonstrate that dietary protein affects intestinal calcium absorption. We conducted a series of short-term nutrition intervention trials in healthy adults where dietary protein was adjusted to either low, medium or high. The highest protein diet resulted in hypercalciuria with no change in serum parathyroid hormone. Surprisingly, within 4 d, the low protein diet induced secondary hyperparathyroidism that persisted for 2 wk. The secondary hyperparathyroidism induced by the low protein diet was attributed to a reduction in intestinal calcium absorption (as assessed by dual stable calcium isotopes). The long-term consequences of these low protein-induced changes in calcium metabolism are not known, but they could be detrimental to skeletal health. Several recent epidemiological studies demonstrate reduced bone density and increased rates of bone loss in individuals habitually consuming low protein diets. Therefore, studies are needed to determine whether low protein intakes directly affect rates of bone resorption, bone formation or both.